

Fit for the Engine? An Audit of Image-to-3D Meshes in a Completed Unity Game

Conventions, Texture Memory, Simplification and Animated Contact

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Abstract

Image-to-3D generators let a solo developer fill a game with textured models made from single drawings. We audited the 26 generated meshes of a completed Unity game (24 static models from one Hunyuan 3D setting, a companion we simplified, and a rigged, animated boss) against a plan committed before any audit script ran, then extended the audit after an internal review. Three of five stated hypotheses were not supported. The raw meshes are closed and edge-manifold, and their UV layouts fall within the range of the game’s purchased meshes. Most problems came from the pipeline around the generator. The importer reversed the characters’ front, which glTF places at +Z. The import path stored each asset’s 4096² base map uncompressed (21 of 24 with a CPU copy) and dropped its normal and metallic-roughness maps; these 24 maps are 70 % of the texture data in the build, and none of it left a trace in the commit history. Our simplification of the companion tore it into 2,433 pieces because the simplifier (MeshLab) was given the seam-split vertex buffer its glTF loader returns; welding first leaves every asset crack-free. Two problems did come from the generator: every asset is scaled to about one metre, and the rigged boss’s idle animation lifts it 0.46–0.68 m above where its bind pose says its feet are. Simplifying in MeshLab the seam-split vertex buffer its glTF loader returns tore the companion into 2,433 pieces; welding first left every asset crack-free at a modest colour cost. Two problems did come from the generator: every asset is scaled to about one metre, and the boss’s idle animation lifts it 0.46–0.68 m above its bind-pose feet, leaving a 0.22 m bob that no constant offset removes. We release scripts, measurements, labelled commit history and review reports, and give a short pre-import checklist derived from the case.

Keywords

image-to-3D generation, game assets, mesh simplification, real-time rendering, asset pipelines, content audit

1 Introduction

Image-to-3D generators now turn a single drawing into a textured mesh in a minute or two [40, 45]. For a solo developer this changes what is affordable: the 15 non-player characters and nine props of the game studied here each came from one drawing. Generators are usually evaluated on how their output looks. Engine readiness depends on different properties: size and orientation, origin, closed and manifold surfaces, memory, whether the mesh survives simplification, and, once rigged, where the feet are when it moves.

This paper audits the generated meshes in a completed Unity game, ChronoTraveler, against those properties, and asks which problems could have been found before import. The analysis plan was committed before any audit script existed; an internal fact-check and review of the first draft then led to further measurements, which we mark as such. The corpus is small (24 assets from one generator setting, a simplified companion and a rigged boss), so this is a case study.

Three of the five stated hypotheses were not supported. The raw meshes are closed and edge-manifold, and their UV layouts fall within the range of the meshes the game bought. Most of what went wrong came from the pipeline around the generator rather than from it. The importer reversed the characters’ front. The import path stored each asset’s 4096² base map uncompressed, in most cases with a CPU copy, and dropped the normal and metallic-roughness maps; these 24 maps are 70 % of the texture data in the build, and none of it left a trace in the commit history. Our simplification of the companion tore it into 2,433 pieces because the simplifier (MeshLab) was given the seam-split vertex buffer its glTF loader returns; welding first leaves every asset crack-free. Two problems did come from the generator: every asset is scaled to about one metre, and the rigged boss’s idle animation lifts it 0.46–0.68 m above where its bind pose says its feet are.

Contributions: (1) an audit of generated meshes in a completed game, with scripts, measurements, commit labels and review reports released; (2) evidence from this case that engine readiness was decided by conventions, the import path, simplification input and animation rather than by mesh topology; (3) a short pre-import checklist (Section 6), derived here and not yet validated elsewhere. It extends our earlier audits of AI-generated game content [43, 44] from text to geometry.

2 Related Work

Image-to-3D generators. Feed-forward and diffusion systems produce textured meshes from one image in seconds to minutes: TripoSR [31], InstantMesh [41], SF3D [4], TRELLIS [40] and the Hunyuan3D series [27, 45], whose commercial service made our assets [28]. Several properties we measure follow from such designs. Hunyuan3D 2.1 normalises training shapes to a unit cube and extracts surfaces as isosurfaces [27], which predicts unit scale and closed meshes; its open-source texturing defaults to 4096² maps and unwraps with default xatlas settings [29], and its developers note that xatlas “generates over-fragmented cuts” [19]. We cannot confirm that the commercial service runs the same code.

Generated assets in production. Evaluation of generated 3D is mostly image-based: T3Bench [12], GPTEval3D [39] and 3DGenBench [42] score rendered views. Work on artist-like topology [5, 26] changes the output representation instead. A 2026 survey of production-ready generation observes that packing efficiency, manifoldness and watertightness are “rarely reported” or “almost never quantified” [38]. PartUV measures chart counts on 114 TRELLIS meshes (xatlas: median 895, mean 1,542) [37]. The closest study to ours is Begemann and Hutson’s practice-led comparison, which

rebuilt one game scene with Meshy and Hunyuan 3D and reported fragmented UV maps and loss of texture integrity under decimation [2]. Practitioners describe generated UVs as “a jumbled mess” [8], and HCI studies document how solo and indie developers use generative tools [24]. Our study differs in measuring every generated asset of a completed game, per asset, against stated hypotheses, and in tracing each problem to where in the pipeline it arose.

Conventions and import. glTF fixes units (metres), up (+Y) and front (+Z), and requires a separate vertex wherever an attribute is discontinuous [30]; Khronos’s importer for Blender offers to merge such vertices only as an option [17]. Khronos’s real-time asset guidelines ask for 1:1 real-world scale and an origin at the bottom centre [16], and engine vendors note that textures usually dominate memory [32]. Several items of our checklist therefore restate existing guidance; what this case adds is which violations occurred, where they came from and which ones went unnoticed.

Simplification with seams. Quadric error simplification [10] was extended to appearance attributes [11, 14]; Hoppe’s wedge structure stores one vertex with several attribute values at a seam [14]. Texture-aware simplification bounds texture stretch [25], seam-aware methods preserve the parametrisation exactly [21], and recent work simplifies textured meshes independently of their UV layout [20]. meshoptimizer’s documentation states that “identical vertices” must be welded before simplification [15]; PyMeshLab’s documentation says its per-wedge conversion “does not merge superfluous vertices” [36]; a simplifier evaluated on Hunyuan3D output merges coincident vertices as standard pre-processing [3]; and an open-source game project hit the same cracking on Meshy assets and also welded first (then re-unwrapped and re-baked), two weeks before we wrote this [9]. The mechanism is known; we report how much damage it did in a shipped asset and that the filter gave no sign of it.

Grounding animated characters. Keeping feet on the ground is a per-frame problem, handled by foot-contact cleanup or inverse kinematics [18]. Unity offers a per-frame “Feet” root mode that “prevents [the] floating problem”, but only for Humanoid rigs [35]; our boss is a Generic rig. Unity precomputes a skinned mesh’s bounds from its animations unless asked to update them every frame [33], and its pose snapshot API has a scale option whose effect depends on the hierarchy [34].

3 The Game and Its Generated Assets

ChronoTraveler is a turn-based educational RPG built in Unity 6 (URP) by one developer working with an AI coding assistant in mid-2026 (repository history June–August), as an MSc project. We call its final build (game commit 87fdd16) the *completed* game; it was not released commercially. Its five historical eras each have three non-player characters (NPCs); the Silk Road era also has nine props (two camels, a horse, a cart, a stupa, a milestone, silk bolts and spice sacks). All 24 were generated from one front-view drawing each with Tencent’s Hunyuan 3D web service (image-to-3D, version 3.1 according to the game’s pipeline notes, “50k faces” setting) and downloaded as GLB. Two more meshes came from the same service: the companion fox, generated at a higher

Table 1: Audited meshes. “Raw” = generator output as downloaded.

Group	<i>n</i>	Source and use
NPCs	15	image-to-3D, 50k setting; raw GLB, static
Props	9	same pipeline; raw GLB, static
Companion (raw)	1	image-to-3D, high setting; 497,850 triangles; not shipped
Companion (shipped)	1	our simplification of the raw mesh; 82,829 triangles
Boss	1	studio pipeline: retopology, rig, 4 animations (4 FBX)

setting (497,850 triangles) and simplified by us to 82,829 triangles before shipping, and the final boss, produced by the service’s studio pipeline [19] with retopology, UV unwrapping, texturing, a 28-bone rig and four animations, delivered as four FBX files that share one 9,038-triangle mesh. We audit 26 meshes, of which 25 shipped (Table 1); `data/manifest.json` lists every file with its SHA-256.

The 24 GLBs in the game are byte-identical to the downloads. They were imported with UniGLTF 0.131 (the glTF importer bundled with UniVRM), whose default converts glTF to Unity coordinates by reversing Z. An editor tool then rescaled each NPC to 1.7 m, turned it 180° and replaced its material with URP/Lit; a second tool does the same for the boss at 10 m; a run-time component (GroundSnap) moves the animated boss down onto its platform.

4 Method

The audit plan—metrics, five hypotheses and decision rules—was committed and pushed to the study repository (commit 2b0bc54, 16:32 on the day of the audit) before any audit script existed. It is not a registered report: the plan was written by the same person who built the game and wrote the commits that B5 labels, so B5 and E1 are not blind. Two of the five hypotheses restate facts a read-only survey had already seen while scoping the study (triangle counts, texture sizes, the double-sided flag, the root rotation, unit-scale heights); the plan says so. The scripts and all first results were then committed together 21 minutes later, so the order in which they ran is not separately recorded. Everything added after the first draft, in response to an internal fact-check and review, is marked as such and listed in Section 8.

Terms. glTF stores a *seam-split* vertex buffer: wherever a per-vertex attribute such as a UV coordinate is discontinuous, the vertex is duplicated. *Welding* merges vertices with identical positions and keeps UVs per corner, as *wedge* attributes [14]. After welding, a mesh is *closed* if it has no boundary edges and *edge-manifold* if every edge has exactly two incident triangles. MiB is 2²⁰ bytes.

B1 Conventions. A small GLB reader (no glTF library, so the numbers come from the bytes any importer reads) applies each file’s node chain and records the bounding box, the largest extent, the origin as a fraction of the box per axis, node transforms, material flags and extensions, and each embedded image’s format and size. **H1:** every raw asset is scale-normalised (largest extent 0.9–1.2 units), carries a non-identity root rotation and is double-sided. *Added after*

review: each GLB was rendered in glTF coordinates from +Z and -Z to see which side is the front.

B2 Hygiene. After welding: connected components (triangles sharing a vertex), boundary and non-manifold edges, non-manifold vertices, degenerate and duplicate triangles, and self-intersecting triangles (MeshLab [6] via PyMeshLab [23]). Before welding: UV charts (seams separate them) and UV coverage, the share of the unit square covered by UV triangles rasterised at 2048^2 . Rasterisation counts edge texels, so coverage is an upper bound by about two points; chart overlap was not measured. **H2:** a majority of the 24 raw assets fail at least one of closed, edge-manifold, or single component. *Added after review:* the same measurements on the game’s 81 non-generated meshes (a licensed character and three purchased environment packs).

B3 Cost. Geometry bytes as stored and texture bytes for three GPU formats with full mip chains: RGBA8 (4 B/texel), BC7 (1 B/texel), BC1 (0.5 B/texel). **H3:** texture memory exceeds geometry memory at least tenfold for every raw asset in every format. *Added after review:* the textures actually stored in the shipped Windows build, read with UnityPy.

B4 Simplification. Texture-aware quadric edge collapse [10, 11, 14] as implemented in MeshLab (quality threshold 0.3, boundary preservation off) at 50, 25, 12.5 and 6.25 % of the triangles, up to three passes. The plan named a with-UV and a no-UV condition. Because B2 found the shipped companion torn, the with-UV condition was run on two inputs: *S*, the seam-split buffer that MeshLab’s glTF loader returns (what our companion pipeline used), and *W*, the same file after `meshing_remove_duplicate_vertices`. *N* is welded, no UVs. We record triangles reached, boundary edges in the output after welding, and surface distance to the original sampled Metro-style [7] (200,000 samples per direction; the larger of the two directions, as a percentage of the bounding-box diagonal; we report the mean and the maximum). **H4:** with UVs preserved, most assets cannot reach 6.25 % within three passes while the no-UV control can. *Added after review* (12.5 and 6.25 %): the same *S* and *W* inputs in `meshoptimizer` [15]; *S* with boundary preservation on; *N* with the same normal flag as *S* and *W*; and a colour error for *S* and *W*, comparing base colour looked up through the simplified UVs with base colour at the nearest original vertex (60,000 samples; the same procedure on the unsimplified mesh gives the noise floor).

B5 History. All 484 non-merge commits up to the final build were searched with a pre-stated pattern of 25 terms (scale, pivot, ground, clip, cull, bounds, glb, decimat, ...). Each of the 156 matches was labelled ASSET (root cause is a property of a generated asset), ENGINE (an engine or tool behaviour any asset would hit), LEVEL (level geometry, placement, purchased or procedural content) or OTHER, with ambiguous cases sent to OTHER. Two AI labellers drafted the labels, each taking half; the AI assistant reviewed every ASSET, ENGINE and low-confidence label in two rounds, recording each change with its reason, and the author signed off. For ASSET commits we note which B1–B3 measurement, if any, would have shown the property before import. **H5:** most ASSET fixes concern properties such counting would reveal.

Table 2: Hypotheses as stated in the plan and their outcome.

	Statement (abridged)	Outcome
H1	all raw assets scale-normalised (0.9–1.2), rotated root, double-sided	not met (23/24)
H2	most raw assets not closed, not manifold, or several components	not supported (5/24)
H3	texture $\geq 10\times$ geometry memory, all assets, all formats	supported (24/24)
H4	with UVs, most assets cannot reach 6.25 %	not supported (S: 6/24, W: 1/24)
H5	most ASSET fixes visible to counting	supported only after relabelling

E1 Grounding (exploratory). In Blender 5.1 we evaluate the boss’s skinned mesh at every frame of each animation (60 fps) and take the lowest vertex as ground truth. Heights are scaled so that the bind pose is 10 m tall, as in the game. We score the estimators the game tried: bind-pose bounds, one frame, and the shipped minimum over ten samples 0.16 s apart, at every starting phase of the loop.

5 Results

Table 2 summarises the five hypotheses. The two that were met, or nearly met, restate what the scoping survey had already seen; none of the three that had not been looked at held as first analysed.

5.1 Conventions: size from outside, front by the importer

All 24 raw assets carry the same root rotation (+90° about *X*, so the mesh is stored with its up axis along local -Z), all are double-sided, and all use the same material: one primitive, one material with `KHR_materials_specular`, and three 4096² PNG maps (base colour, normal, metallic-roughness). Every file names Blender’s glTF exporter as its generator. Scale is normalised whatever the subject: the largest extent is 0.895–1.151 units (median 1.082), and glTF units are metres [30], so a soldier, a camel and a stupa all arrive about one metre long. The milestone falls 0.005 below the plan’s 0.9 bound, so H1 as written is not met. The origin sits on the base of the bounding box in every case and within 0.44–0.58 of its width and 0.30–0.69 of its depth.

Rendered in glTF coordinates, all 15 NPCs show their faces from +Z, the front the glTF specification prescribes (Figure 1, top). They faced away from the player in the game because UniGLTF’s default conversion reverses *Z*, and the game’s NPC code turns an object’s +Z toward the player. The 180° turn in the import tool compensated for the importer, not for the generator. Real-world size, by contrast, cannot be recovered from the file: it has to come from the design.

5.2 Hygiene: closed and manifold; UV layouts like purchased meshes

H2 is not supported. After welding, all 24 raw assets are closed, edge-manifold and vertex-manifold, with no degenerate or duplicate triangles; 19 are one connected component and five have two to four. Every one has self-intersecting triangles (11–524, median 64).



Figure 1: Top: two raw NPCs rendered in glTF coordinates from +Z and -Z; the faces are on the +Z side, as glTF specifies. Bottom: close-up of the companion at the shipped 82,829 triangles. Simplified from the seam-split buffer (as shipped) the surface is cracked and the orange background shows through; simplified after welding it is intact.

Closed, manifold output is what isosurface extraction produces [27], so in hindsight H2 was the wrong expectation. Other studies describe generated meshes, including Hunyuan3D 2.0 output, as noisy or non-manifold [3, 37]; we measure after welding, which may explain part of the difference.

Nor do the UV layouts stand out. The raw assets have 703–3,088 UV charts (14–63 per thousand triangles, median 32), their charts cover 48–68 % of the texture square (median 53 %), and seams duplicate 1.31–1.81 vertices per welded vertex. Our chart counts sit in the range PartUV reports for xatlas on generated meshes [37]. The game’s 81 non-generated meshes span similar ranges: the licensed player character has 10 charts per thousand triangles (its charts are stacked, so coverage is not comparable); the purchased Temple of Nike pack has 10–44 charts per thousand and covers 36–99 %; the Altar Ruins pack covers a median of 14 %; the Egypt pack maps most surfaces to a colour palette and covers almost nothing. Self-intersections are common in the non-generated meshes too (54 of 81), and half of them are not closed (41 of 81). By these measures the generated meshes are as tidy as what the game bought.

The one generated mesh that fails every check is the one we processed. The raw companion is closed, manifold and one component; the shipped version has 44,316 boundary edges, 1,442 non-manifold vertices, 2,433 components and 9,868 self-intersecting triangles (19 before). Section 5.4 shows why.

5.3 Cost: one uncompressed map per asset, 70 % of the build’s textures

H3 holds for all 24 assets: geometry occupies 1.28–1.60 MiB as stored, while three 4096² maps would take 256 MiB uncompressed, 64 MiB in BC7 and 32 MiB in BC1 with mips, 20–25 times the geometry even in the smallest format [1, 22]. It fails for the companion’s high-setting original (497,850 triangles), where the BC7 ratio is 4.3.

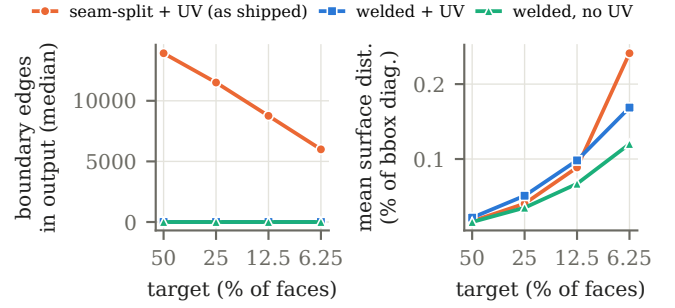


Figure 2: Simplifying the 24 raw assets (medians). Left: boundary edges in the output, counted after welding identical positions; the inputs have none. Seam-split input cracks at every target; welded input, with or without UVs, never does (the two lines coincide at zero). Right: mean surface distance to the original.

Table 3: Simplification at 6.25 % of the triangles (24 assets, up to three passes). Distance: mean surface distance, % of the bounding-box diagonal, median [max]. Colour: base-colour RMSE above the unsimplified noise floor (0–255 scale), median [max]. The maxima for W both come from the asset that stalled. *Normal preservation on; with it off, as for S and W, all 24 also reach the target with no cracks.

Input	Reached	Cracked	Distance	Colour
S: seam-split + UV	18	24	0.24 [0.32]	3.5 [9.6]
S, boundary preserved	0	0	–	2.4 [16.7]
W: welded + UV	23	0	0.17 [1.40]	3.7 [37.1]
N: welded, no UV*	24	0	0.12 [0.20]	–

The shipped build shows what the pipeline actually did with those maps. Each of the 24 assets kept one texture, its base colour: the import tool’s material conversion copied the main texture and left the normal and metallic-roughness slots empty. That one map is stored at 4096² as uncompressed ARGB32 with a full mip chain, 85.3 MiB, because UniGLTF creates embedded textures with LoadImage rather than through Unity’s texture importer; 21 of the 24 are also flagged CPU-readable, which keeps a second copy in system memory. Together the 24 maps are 2,048 MiB, 70 % of the 2,944 MiB of texture data in the build; the largest texture from any other source is 21.3 MiB (a 4096² BC7 map). The boss’s and companion’s textures, which did go through the texture importer, were stored block-compressed (DXT1 for colour, DXT5 for normals); at DXT1 a 4096² base map would be 10.7 MiB, at BC7 21.3 MiB. None of this appears in the game’s commit history.

5.4 Simplification: the input decides, not the UVs

H4 is not supported. Welded and with UVs preserved, 23 of 24 assets reached 6.25 % of their triangles (about 3,100) in one pass with no cracks (Table 3). The exception has the most UV charts (3,088) and stalled at 3,800–3,900 triangles after three passes (two runs). Charts

constrain where the simplifier may collapse, but at these budgets they rarely stop it.

What decided the outcome was the input. MeshLab’s glTF loader returns the file’s seam-split buffer, in which every UV seam already separates the surface into distinct vertices, as glTF requires [30]. MeshLab’s texture-aware simplification then treats each seam as an open boundary. With boundary preservation off (the default) the two sides of every seam move independently: every asset cracked at every target, with a median of 13,924 open edges already at 50 % (Figure 2), and six assets did not reach 6.25 % within three passes. With boundary preservation on, nothing cracks but nothing gets far: no asset went below 13,238 triangles, and the median stopped at 21,939, about 44 % of the original. The filter reports no error in either case. The failure depends on the tool. meshoptimizer, which infers seams from matching positions [15], did not crack the same seam-split inputs but stalled: two of 24 reached 6.25 % (median 5,900 triangles), against 24 of 24 on welded input.

Welding identical positions first removes the problem in both tools; in PyMeshLab this is one call, `meshing_remove_duplicate_vertices`, which keeps UVs per corner. The cost in colour is modest: median base-colour error above the noise floor is 1.5 levels at 12.5 % (0.1 for seam-split input) and 3.7 at 6.25 % (3.5), on a 0–255 scale, with one outlier at 37 for the asset that stalled. Mean surface distance is slightly higher than for seam-split input at 50–12.5 % and lower at 6.25 %. Attribute discontinuities are a classic concern in simplification [13, 14], and the weld-first rule is documented [15]; what this case shows is how silently a common scripted path runs into them.

This is what happened to the companion. Our pipeline loaded the 497,850-triangle GLB directly and simplified it in three passes to 82,829 triangles, giving the torn mesh of Section 5.2 (Figure 1, bottom). The design note for that step says the simplifier stopped near 106k triangles because the UV islands were a hard floor. That figure came from an earlier trial run; the shipped script multiplies the target by 0.55 per pass and simply ran its three passes ($497,850 \times 0.55^3 \approx 82,830$). On the welded input the same filter takes the companion to 31,116 triangles (6.25 %) in one pass with no boundary edges. The torn mesh shipped after a play check recorded only as “Play-verified” in the commit history.

5.5 History: few fixes, and most of them ours

Of 484 non-merge commits, 156 matched the pattern. After review they are 61 LEVEL, 78 OTHER, 7 ENGINE and 10 ASSET. The pattern was written around asset terms, so the comparison is within matched commits only. Seven of the ten ASSET commits are tooling written ahead of need: the NPC import tool (scale to 1.7 m, turn 180°, convert material), its application to each era’s models, the boss bake tool, and the companion’s simplification and pivot setup. Three are fixes: the boss hovering above its platform (Section 5.6), and twice the companion lying on its back after our GLB-to-OBJ conversion dropped the root rotation node.

The first labelling had six ASSET fixes and found H5 not supported (three of six countable). Review moved the NPC facing fix to ENGINE (Section 5.1) and two ambiguous fixes to OTHER, as the plan’s rule requires: a double 180° turn in our own boss code, and a scale calibration whose cause mixed the rig and the engine. The

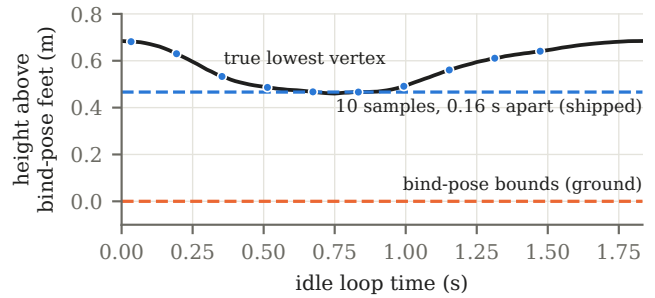


Figure 3: Boss idle loop (1.85 s) at game scale. Placing the bind-pose feet on the ground leaves the animated feet 0.46–0.68 m in the air for the whole loop. The shipped estimator (minimum of ten samples, dots) lands within about 20 mm of the loop’s minimum at any starting phase; the remaining 0.22 m is the idle bob.

boss-hover fix was also labelled with low confidence, but its cause is not in doubt (E1 reproduces it offline from the asset alone), so it stays ASSET. Two of the three remaining fixes (the lost rotation node) concern a property counting would show, and so do nine of the ten ASSET commits, so H5 is supported; because the relabelling came after the result, we treat this as weak evidence. Two of the three fixes repaired a step we had added.

5.6 Grounding: the bind pose misplaces an animated character

The boss’s studio mesh is skinned with its bind-pose feet at the origin, but its idle animation lifts the whole body (Figure 3). The bake tool put the bind-pose feet on the platform, which leaves the animated feet 0.46–0.68 m above it on every frame. The first runtime fix read the renderer’s bounds, which in the developer’s test did not follow the pose; Unity documents skinned-mesh bounds as precomputed unless updated every frame [33]. One sampled frame can land anywhere in the 0.22 m bob. The shipped estimator, the minimum of ten samples 0.16 s apart, covers 1.44 s of the 1.85 s loop and in Blender lands within about 20 mm of the loop’s minimum at any starting phase (median 4 mm). The remaining 0.22 m is not estimator error: one constant offset cannot follow a body that moves up and down, so it is the floor for any constant, and only per-frame contact handling such as foot IK [18] goes below it. Unity’s per-frame “Feet” root mode does this for Humanoid rigs only [35]; the generated boss is imported as a Generic rig.

The constant is also calibrated on one clip. The other three animations dip lower than idle: relative to the idle-calibrated ground, the lowest frame of the attack sinks 0.27 m, the knock-down 0.08 m and the forward leap 1.19 m. These are offline predictions; we did not check them in play. In the engine the estimator also needed an empirical 0.30 m offset, attributed in the commit record to a scale ambiguity in Unity’s BakeMesh for this rig; we did not reproduce or evaluate it.

6 Discussion

Where the problems were. We expected generated meshes to fail as meshes. They did not: mesh topology caused no fix, and by our measures the raw output is as closed, as manifold and as efficiently unwrapped as the meshes the game bought. The problems sat around the mesh. The generator normalises size to one metre whatever the subject. The importer reversed the front. The import path stored each asset’s texture uncompressed, mostly with a CPU copy, and dropped two of its three maps. Our simplification tore the companion, and the “UV floor” recorded for it is not reproduced by our runs. An animation lifted the boss off the ground. Of these, only the size normalisation and the animation belong to the generator; the rest are properties of the pipeline we built around it, and none of the texture problems left a trace in the commit history.

A pre-import checklist. Derived from this case and not validated on other projects.

- (1) *Size and front.* Treat the largest extent as unknown and supply real size from the design. Render one view from +Z before import and one after, to separate the asset’s front from the importer’s axis convention.
- (2) *What the engine actually stores.* Read texture format, size, readability and which maps the material references from the build, not from the source file; an uncompressed 4096^2 map is four times its BC7 size, and a dropped normal map is invisible to a memory check.
- (3) *Simplification input.* Weld before simplifying and count boundary edges after; a seam-split buffer simplifies without any error message, and in MeshLab usually returns the requested triangle count.
- (4) *Animated contact.* For rigged assets, evaluate every clip offline and compare its lowest vertex with the bind pose and with the clip used for placement.
- (5) *Topology.* Closedness and manifoldness were never the problem here, but they are cheap to count, and our own tool was the one that broke them.

Items 1 and 3 restate existing guidance on real-world scale and origin [16] and on welding before simplification [15], and texture memory is a familiar concern [32]; what this case adds is where the violations actually came from and which ones a developer did not notice.

Counting and looking. Our earlier audit of an LLM-generated quiz bank found that the defects that mattered were invisible one item at a time and obvious when counted across the set [43]. Here counting found the uniform size convention, the torn companion and the texture budget, but two findings needed a picture or a simulation: the front, which took a render in a known coordinate frame, and the hover, which took evaluating the animation. A useful pre-import check for generated geometry includes a small render and a short animation pass, not only statistics.

7 Limitations

Scope. One generator and version, one importer, one game: 24 meshes from one setting and two from other routes. Hunyuan 3D’s service is closed and changes over time. The 50k setting and 4096^2 textures were our defaults, not a sweep. The texture findings depend

on UniGLTF and our material tool; another importer would store the maps differently. We report counts, not population estimates.

No frame-time measurements. Memory figures are texel counts and stored sizes from the build, not profiler readings; we did not measure draw calls, frame time or GPU residency per asset.

Not blind. The author built the game, wrote the commits B5 labels and wrote the plan. The labels were drafted by two AI labellers and reviewed by the AI assistant and the author; there was no independent human rater. Commit messages record what the developer believed at the time. Keyword search misses commits that never used the words.

Grounding. Ground truth comes from Blender’s evaluation of the FBX, not from the engine; the in-engine offset and the predicted sinking in the other clips were not checked in play. The boss is one rig.

Appearance. Visual fidelity to the input image was not scored. The colour error in B4 uses the nearest original vertex and only the base colour map.

Authorship. The game, this audit and this text were made with an AI coding assistant. Plan, scripts, outputs, labels and the review reports that prompted the revisions are public so that the numbers can be checked without trusting the author or the assistant.

8 Deviations from the Plan

- B4’s with-UV condition was split into seam-split (S) and welded (W) inputs after B2 found the shipped companion torn.
- After an internal fact-check and review of the first draft, we added: the facing renders (B1), the non-generated baseline (B2), the shipped-build texture readout (B3), S with boundary preservation, a matched no-UV control and a colour error (B4), all four clips and all starting phases (E1). A scene-memory comparison in the first draft was dropped: its premise (compressed textures) was wrong.
- The first draft stated that UV charts do not overlap; the rasterised coverage cannot show that, and the claim was removed.
- B5 labels were drafted by two labellers rather than one and reviewed twice. The second review moved three fixes out of ASSET, which changed H5’s outcome (Section 5.5). The reviewer added an *origin* field and the fix/feature split, neither of which was in the plan.
- The plan’s B3 metric was bytes as uploaded; we report bytes as stored in the file, then what the build stored.
- E1 did not evaluate the shipped constant offset (0.30 m), which corrects an engine-side reading we did not reproduce. The Wilson interval planned for H2 was dropped: the 24 assets are not a random sample. Output file names differ from those listed in the plan.
- A second internal fact-check of the revised draft led to the meshoptimizer comparison and to corrections of several numbers and phrasings; both review rounds are in the repository.

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